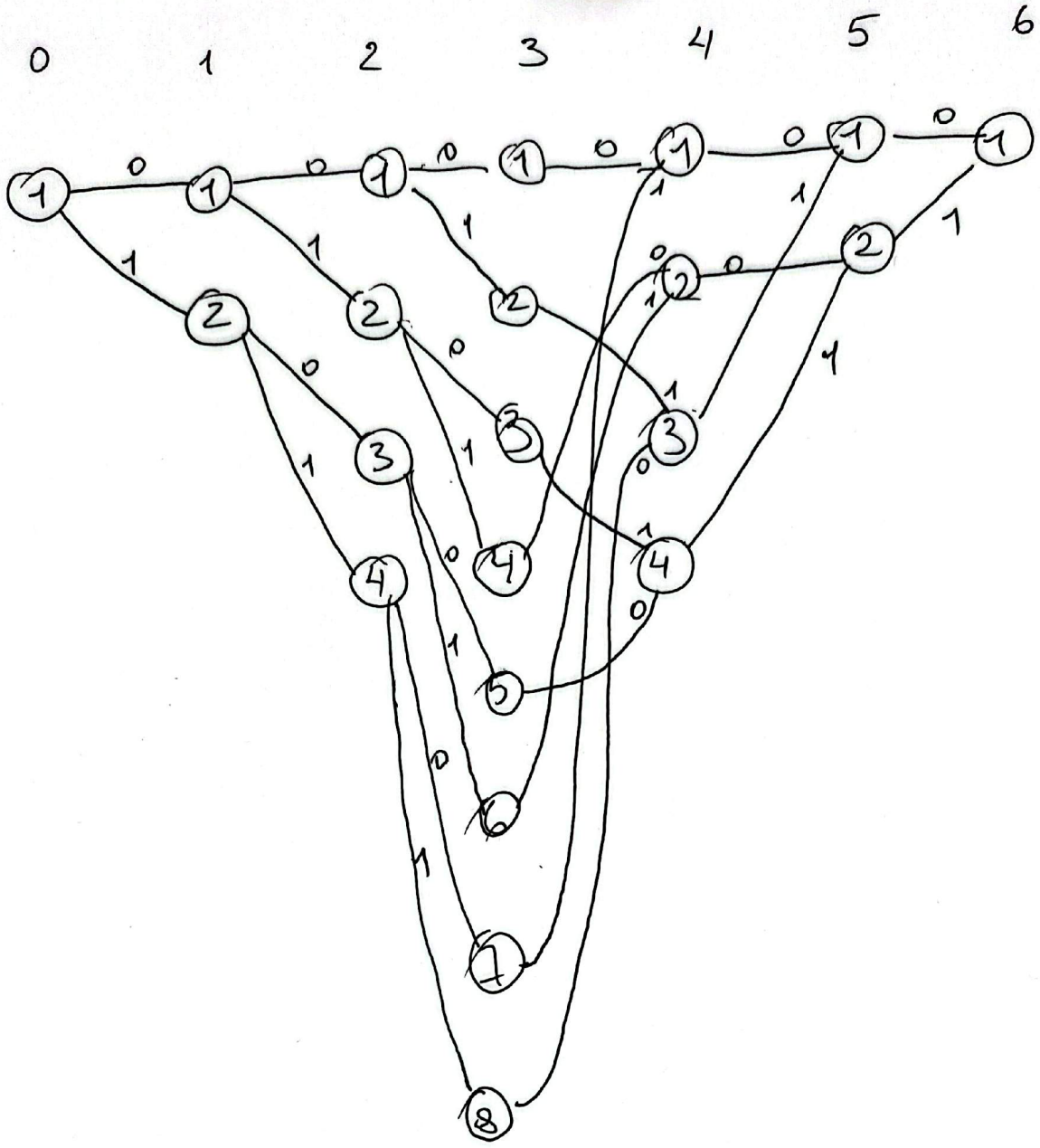
(1)

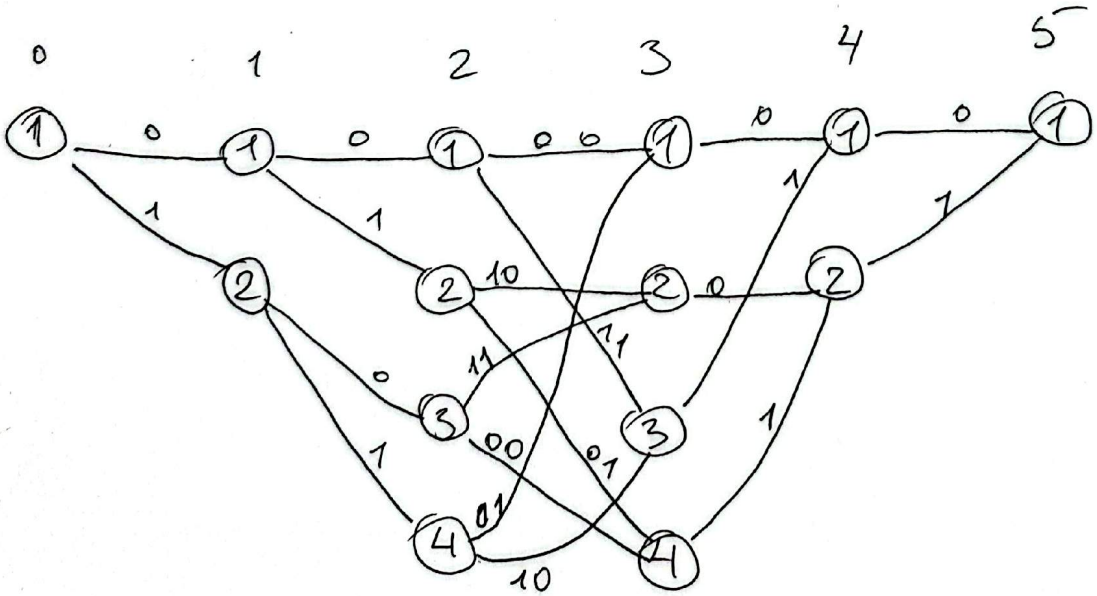
$$G = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 \end{pmatrix}$$

1 кодов

000000
101101
001110
010111
100011
011001
110100
111010

i	$\{c^p\}$	$F_i(c^p)$	$\{V_{i-1}\}$	V_i	c_i
0	$\{\emptyset\}$	$\{c_m\}$	$\emptyset$	1	—
1	0	00000 01110 10111 11001	1	1	0
	1	01101 00011 10100 11010	1	2	1
2	00	0000 1110	1	1	0
	01	0111 1001	1	2	1
	10	1101 0011	2	3	0
	11	0100 1010	2	4	1
3	000 001	000 110	1	1 2	0 1
	010 011	111 001	2	3 4	0 1
	100 101	011 101	3	5 6	0 1
	110 111	100 010	4	7 8	0 1
4	0000, 1101	00	1, 7	1	0, 1
	1011, 0110	01	6, 4	2	1, 0
	0011, 1110	10	2, 8	3	1, 0
	0101, 1000	11	3, 5	4	1, 0
5	00000, 00111, 11010, 11101	0	1, 3, 1, 3	1	0, 1 0, 1
	10110, 01011, 10001, 01100	1	2, 4, 4, 2	2	0, 1 1, 0
6	$\{c\}$	—		1	


0 1 2 3 4 5 6

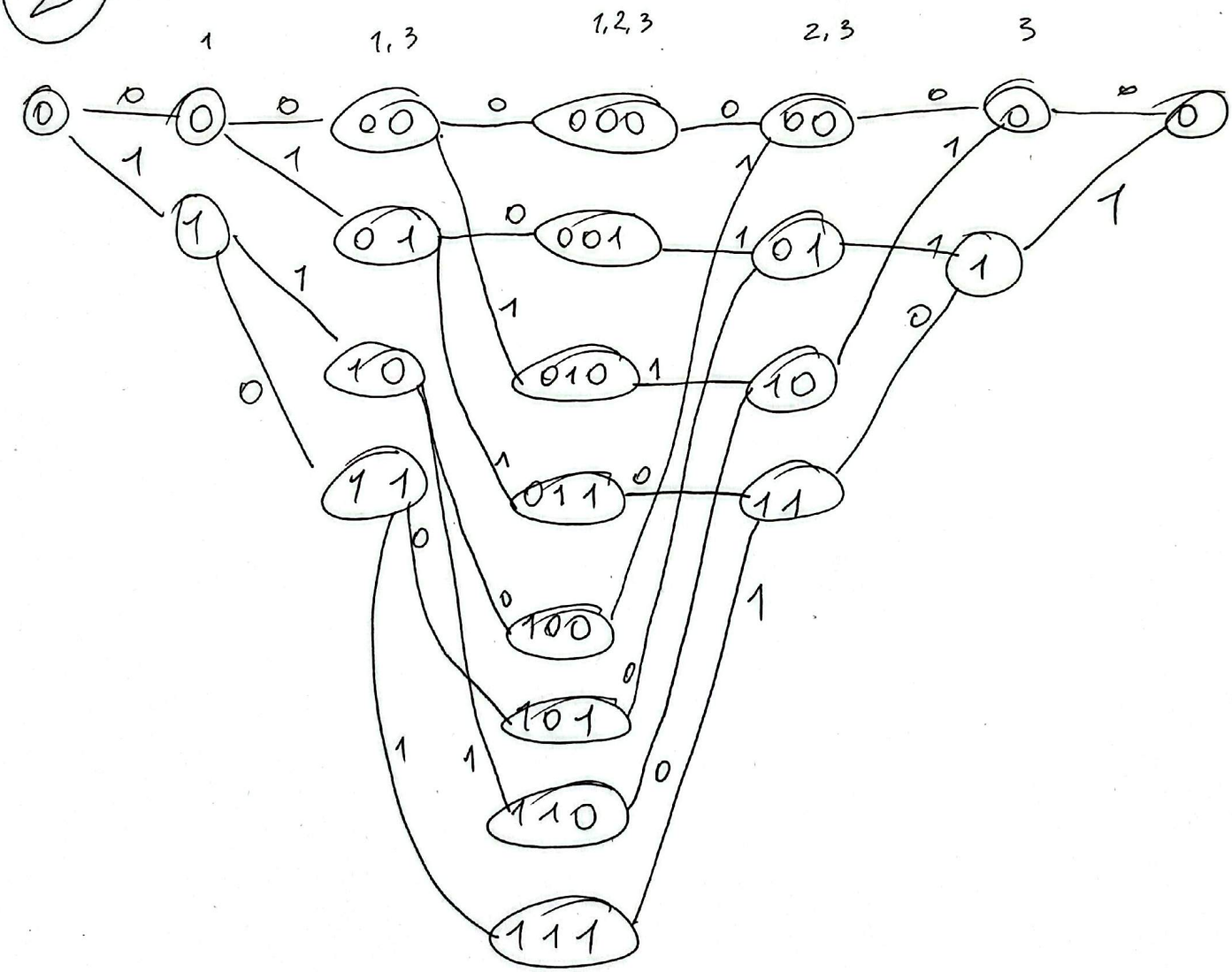

0 1 2 3 4 5

Приведем м-цу G к минимальной сэтовой форме

$$G = \begin{array}{c} \begin{pmatrix} 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 \end{pmatrix} \end{array} \begin{array}{c} g_1 \\ g_2 \\ g_3 \end{array} \xrightarrow{g_1 \pm= g_3} \begin{pmatrix} 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 \end{pmatrix} \xrightarrow{g_1 += g_2}$$

$$\rightsquigarrow \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 \end{pmatrix}$$

2



(3)

$$G = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{pmatrix} = (I \; P)$$

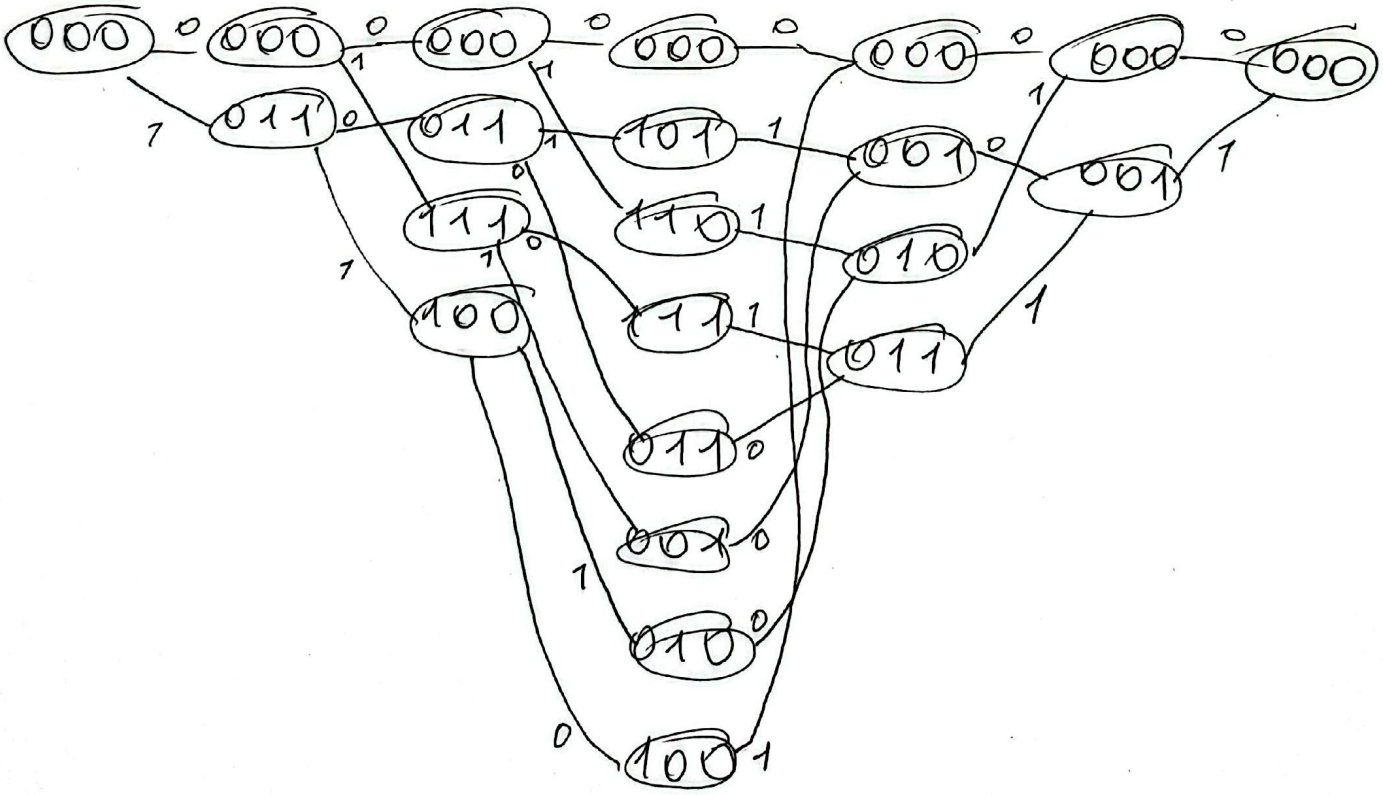
$$H = (P^T \; I) = \begin{pmatrix} 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}$$

i	синдромы
0	000
1	011

00	000
01	011
10	111
11	100

000	000
101	101
001	110
010	111
100	011
011	001
111	010
110	100

0000	000
1011	001
0011	010
0101	011
1000	011
0110	001
1110	010
1101	000

00000	000
10110	001
00111	000
01011	001
10001	001
01100	001
11101	000
11010	000

000000	
101101	
⋮	000
⋮	
KC